

Forecasting Hourly Day-Ahead Electricity Prices in the German-Luxembourg Bidding Zone

Data & AI in Economics - Group T

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Sources:

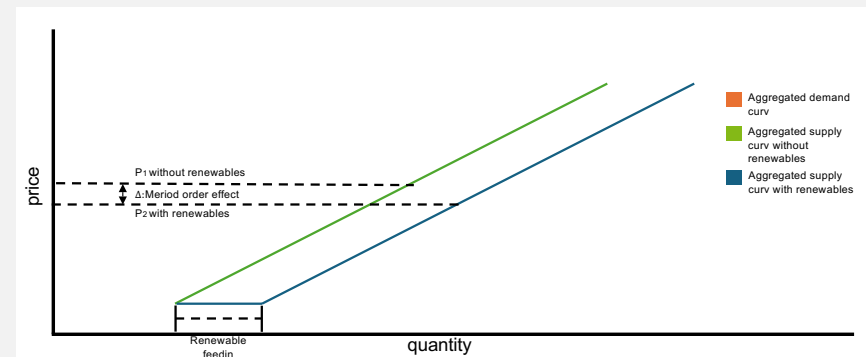
Motivation and economic impact

Day-ahead price forecasting is economically relevant because prices guide short-term electricity market decisions

Why day-ahead electricity price forecasting matters

- Day-ahead electricity prices are economically relevant because they influence short-term decisions of producers, suppliers, traders, industrial consumers and system operators, especially regarding bidding, procurement, generation scheduling and risk management.
- In the German-Luxembourg bidding zone, prices are strongly affected by the expected balance between electricity demand and available supply, making load forecasts and supply forecasts central drivers of hourly price formation.

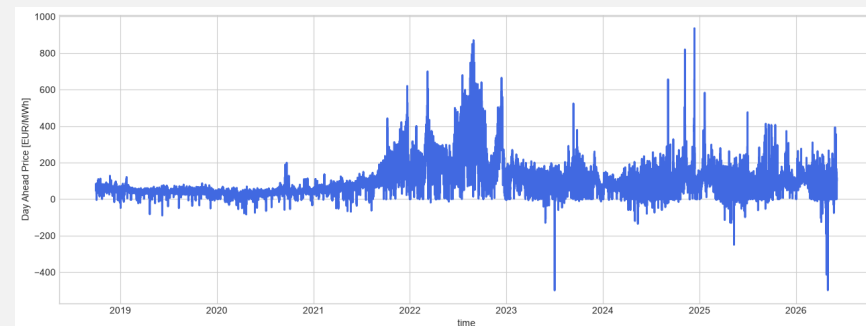
Meriod order effect: the relevance of renewable energy



Research question

How accurately can hourly day-ahead electricity prices in the German-Luxembourg bidding zone be forecasted using market fundamentals, renewable generation forecasts, load forecasts and calendar effects?

Day-ahead electricity price of German-Luxembourg bidding zone



Data

We transformed hourly SMARD market data into a clean and feature-rich forecasting dataset

Data information

- Data provider: SMARD - Strom- und Gasmärkten für Deutschland (Information platform of the Bundesnetzagentur)
- Data range: 01.10.2018 - 04.06.2026
- One row represents one hourly observation on the day-ahead market of the German-Luxembourg bidding zone

Data Preparation

- **Time shifts:** Timestamps are converted into a consistent hourly format. Duplicated hourly observations are aggregated by taking the mean, while missing values are interpolated to preserve a continuous time series.
- **Missing values:** Missing values and non-numeric entries are converted into NaN values. These values are then handled through interpolation or imputation with kNN.
- **Variance stabilization:** The day-ahead price is transformed into a stabilized target variable.
- **Feature Engineering:** Additional features are engineered to capture recurring price patterns. Calendar variables, hourly lag features, and daily minimum/maximum lag features are added to represent seasonality, price persistence, and recent market volatility.

Variable	Type	Role	Description
Timestamp	datetime	Identifier	Start of timeperiod in Central European (Summer-) Time
Total Load FC [MWh]	float	feature	Forecasted total electricity consumption for the following day
Wind Offshore Production FC [MWh]	float	feature	Forecasted net electricity generation from onshore wind turbines for the following day
Wind Onshore Production FC [MWh]	float	feature	Forecasted net electricity generation from onshore wind turbines for the following day.
Photovoltaik Production FC [MWh]	float	feature	Forecasted net electricity generation from photovoltaic systems for the following day. The forecast is part of the SMARD category for forecasted wind and photovoltaic generation.
Other Production FC [MWh]	float	feature	Forecasted net electricity generation from other systems for the following day.
Stabilized Day Ahead Price [EUR/MWh]	float	target	Hourly wholesale electricity price in the day-ahead market with variance stabilization approach applied.
German_holiday	boolean	feature	Indicator for whether the calendar date of the timestamp is a public holiday in Germany.
Luxembourg_holiday	boolean	feature	Indicator for whether the calendar date of the timestamp is a public holiday in Luxembourg.
weekday	categorical	feature	Calendar weekday derived from the timestamp.
hour	categorical	feature	Hour of the day derived from the timestamp.
month	categorical	feature	Calendar month derived from the timestamp.
year	numeric	feature	Calendar year derived from the timestamp.
Lagged Stabilized Day Ahead Price [EUR/MWh] for hour -1,...,-168	float	feature	Hourly wholesale electricity price in the day-ahead market with variance stabilization approach applied for all 24 hours of the last 7 days. This entry corresponds to 168 lag features in total.
Minimum of Lagged Stabilized Day Ahead Price [EUR/MWh] of day -1,...,-7	float	feature	Minimum of hourly wholesale electricity price in the day-ahead market with variance stabilization approach applied for day -1,...,-7. This entry corresponds to 7 lag features in total.
Maximum of Lagged Stabilized Day Ahead Price [EUR/MWh] of day -1,...,-7	float	feature	Maximum of hourly wholesale electricity price in the day-ahead market with variance stabilization approach applied for day -1,...,-7. This entry corresponds to 7 lag features in total.

Causal inference

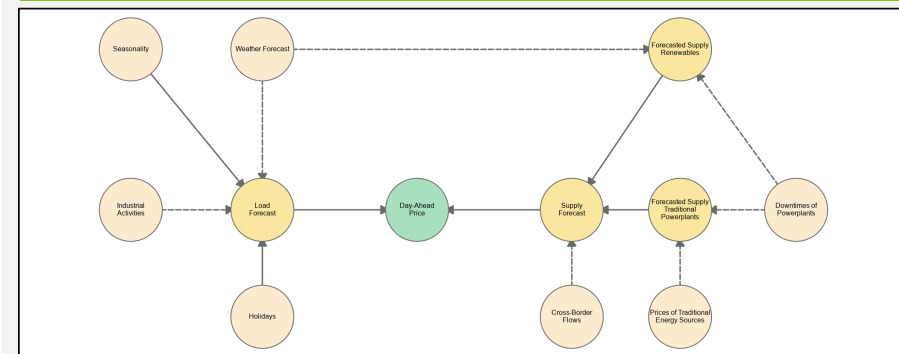
The causal graph structures price drivers and clarifies the limits of causal interpretation

We built a causal graph to structure the assumed economic relationships behind day-ahead price formation. The directed acyclic graph (DAG) connects the day-ahead price to load forecasts and supply forecasts, while renewable generation and traditional production forecasts describe the expected supply side.

Drivers and Backdoor Paths

- **Observed variables:** Holidays, Seasonality, Load Forecast, Supply Forecast, Renewables Supply Forecast, Traditional Powerplant Forecast
 - **Unobserved variables:** Industrial Activities, Weather Forecast, Cross-Border Flows, Downtime of Powerplants, Prices of Traditional Energy Sources
 - **Open back door paths:**
Price $\leftarrow$ Load $\leftarrow$ Weather $\rightarrow$ Renewables $\rightarrow$ Supply $\rightarrow$ Price
Price $\leftarrow$ Load $\leftarrow$ Weather $\rightarrow$ Supply $\rightarrow$ Price
- Due to the open back door paths causal effects can not be estimated

Causal DAG: Drivers of day-ahead electricity prices



Economic interpretation and implication for modelling

- **Economic interpretation:** The DAG reflects the basic market logic of day-ahead price formation through load forecast and supply forecast: high forecasted load or low forecasted supply are expected to increase the day-ahead price, while lower load or higher supply are expected to decrease it.
- **Implication for modelling:** Since the weather forecast is not observed and the relevant backdoor paths remain open, the effect of load and supply forecasts on the day-ahead price cannot be causally identified. Therefore, the supervised learning part focuses on predictive performance and interprets model results as statistical patterns useful for price forecasting.

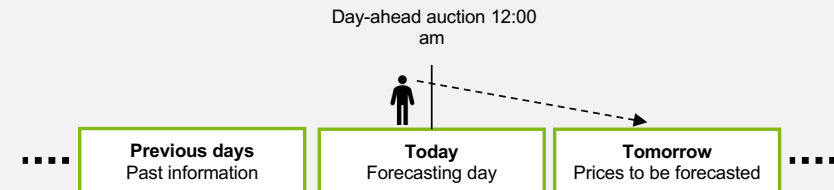
Supervised learning

Random Forest delivers the strongest forecasting performance and clearly improves over both baselines

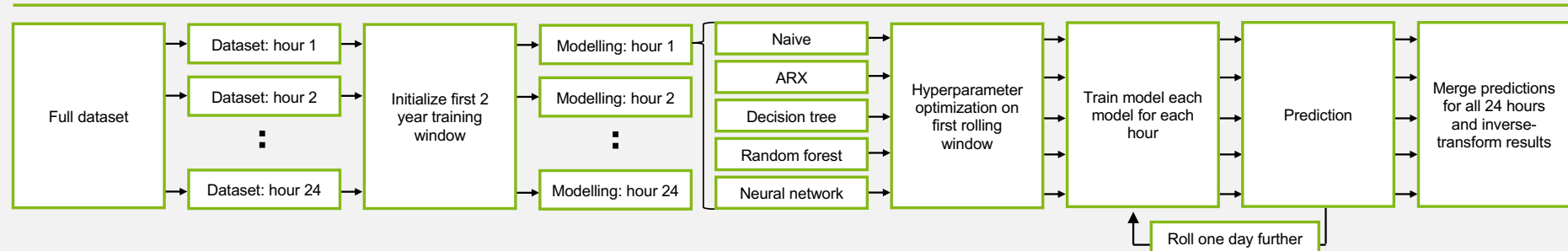
Why day-ahead electricity price forecasting matters

- **Multivariate forecasting approach:** We forecasted day-ahead electricity prices separately for each of the 24 delivery hours using a rolling setup with a two-year training window and one-day-ahead predictions.
- **Model Comparison:** We compared Naive and ARX baselines with Decision Tree, Random Forest and Neural Network models to evaluate whether machine learning adds predictive value beyond simple persistence and autoregressive benchmarks.

Forecasting scenario



Rolling prediction pipeline



Main results

- **Random Forest performs best:** Lowest error with MAE 18.88 €/MWh and RMSE 31.55 €/MWh, reducing MAE by 40.6% compared to the Naive baseline.
- **ML adds predictive value:** Random Forest also outperforms the ARX baseline, suggesting that nonlinear interactions between lagged prices, load, renewables and calendar effects improve forecasting.

Interpretation and limitations

- **Forecast accuracy varies by hour:** Errors are lowest during night and early morning hours and increase during daytime and evening hours, where load variation, renewable feed-in and market activity are more dynamic.
- **Extreme events remain difficult:** The models capture regular price movements well, but large error spikes remain during volatile market phases; therefore, the results improve average forecasting accuracy but do not fully solve price shock prediction.

Implications for clustering

- **Different market states likely matter:** Since forecasting errors vary across hours and market conditions, the following clustering block investigates whether recurring supply-demand regimes help explain different price patterns.
- **Clustering adds an interpretative layer:** Instead of improving the forecast directly, the unsupervised analysis groups similar market situations to better understand different market regimes.

Unsupervised learning

K-Means identifies four interpretable market regimes based on recurring supply-demand situations.

K-Means clustering application

Methodology

- We applied K-Means clustering to identify recurring market regimes in the German-Luxembourg electricity market.
- Clustering was based only on current market-state variables: load forecast, wind offshore/onshore forecasts, photovoltaic forecast and other production forecast.
- Lagged price features and calendar variables were excluded because they were engineered for forecasting rather than for describing market regimes.

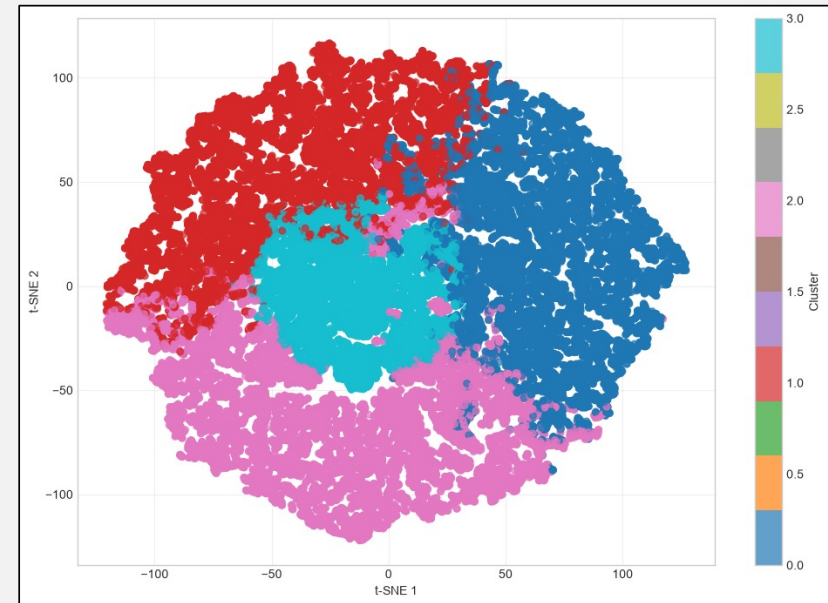
Cluster Selection

- The main clustering solution uses $k = 4$, selected based on the Elbow Method and silhouette score (0,3).
- The t-SNE visualization shows meaningful cluster structure, although the clusters are not perfectly separated.
- The sensitivity analysis supports $k = 4$: $k = 3$ merges relevant regimes, while $k = 5$ mainly splits the wind-related cluster without adding substantial new insight.

Market Regime Interpretation

- The clustering identifies four interpretable supply-demand regimes: low-load, wind-dominated, photovoltaic-dominated and high-load situations.
- The photovoltaic-dominated regime is associated with lower price levels, while the high-load regime is associated with higher prices.
- This supports the idea that day-ahead prices are shaped by different recurring market states rather than one homogeneous market condition.

K-Means cluster visualized by t-SNE



Synthesis & Limitations

- The clustering confirms that the DE-LU day-ahead market is shaped by recurring supply-demand regimes rather than one homogeneous market condition.
- The results are mainly descriptive: K-Means helps interpret market states, but it does not directly improve the forecasting model or provide causal evidence.
- Overall, the clustering adds an interpretative layer to the supervised learning results by explaining why price patterns and forecasting difficulty differ across market situations.

References

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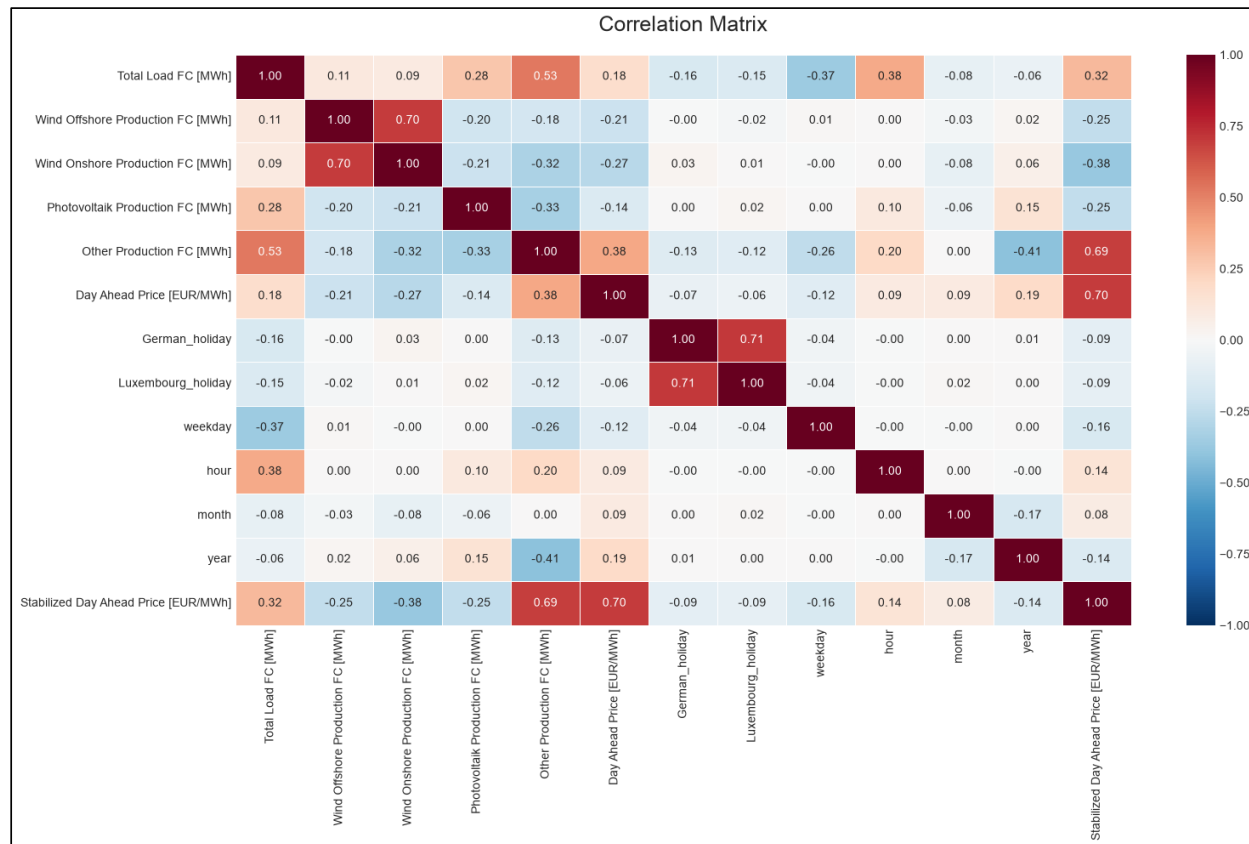
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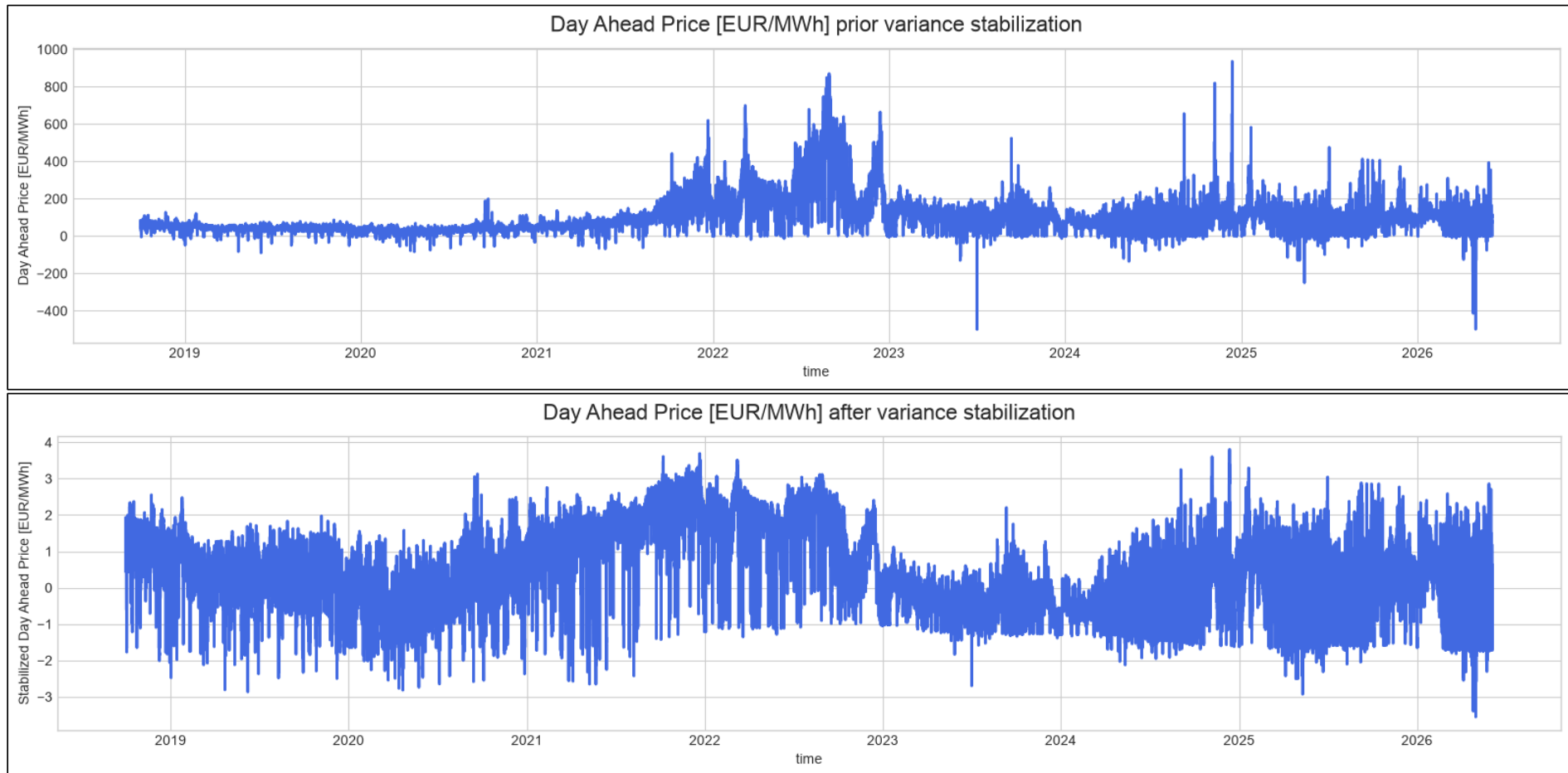
Back Up Graphics

Correlation Matrix

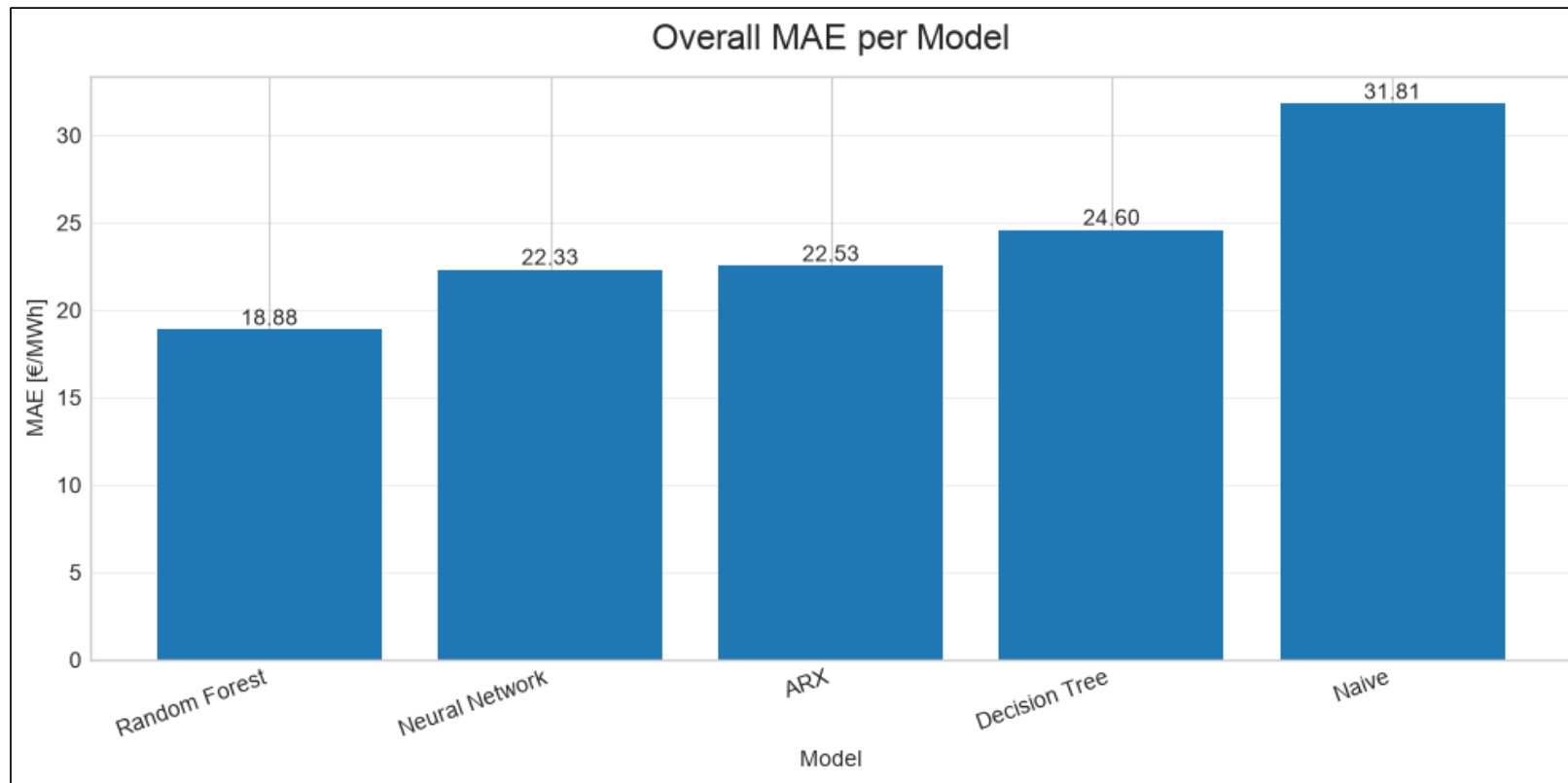


Back Up Graphics

Time series of the day-ahead price before and after variance stabilization

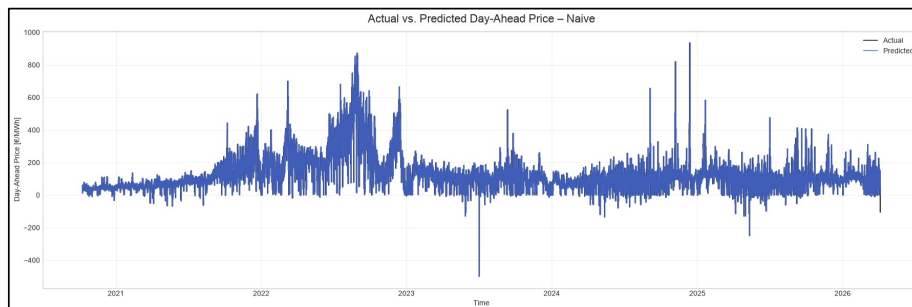
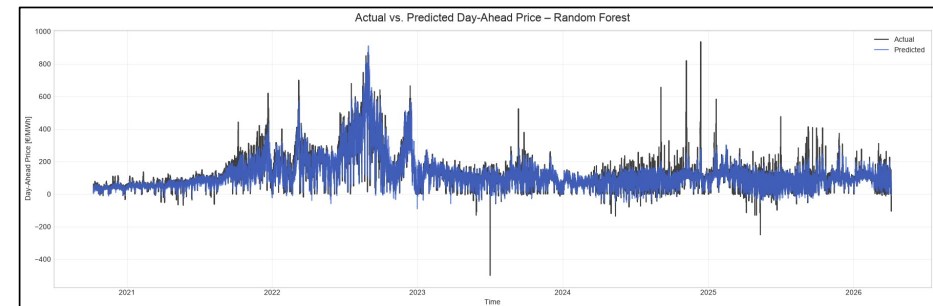
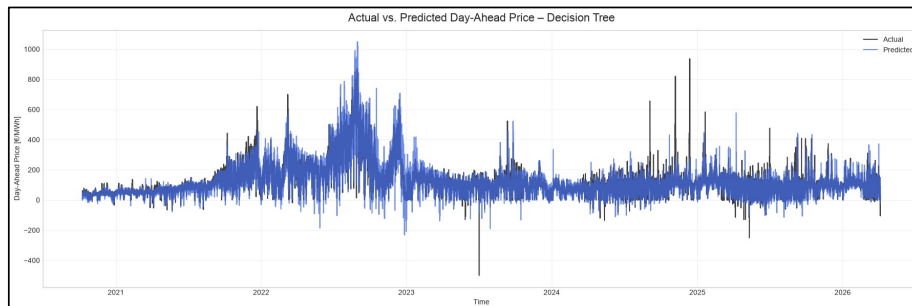
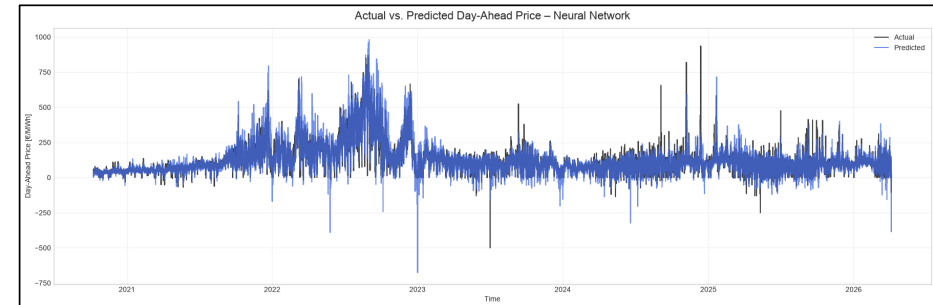
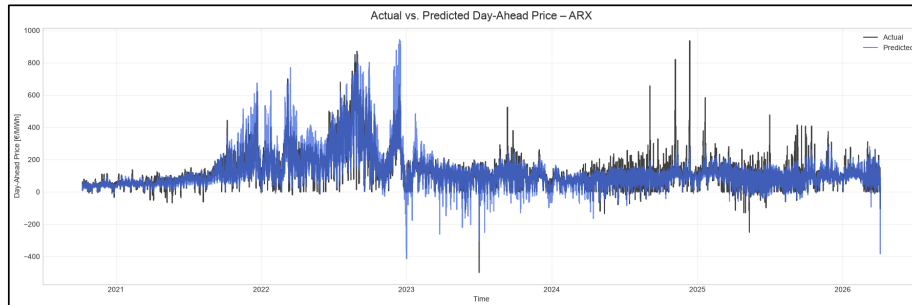


Overall MAE per Model

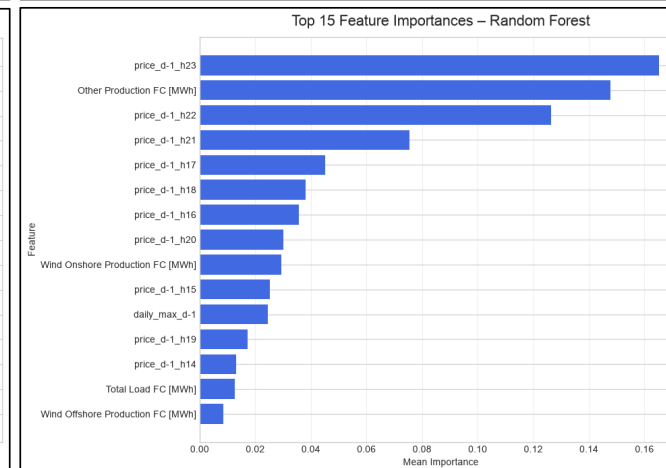
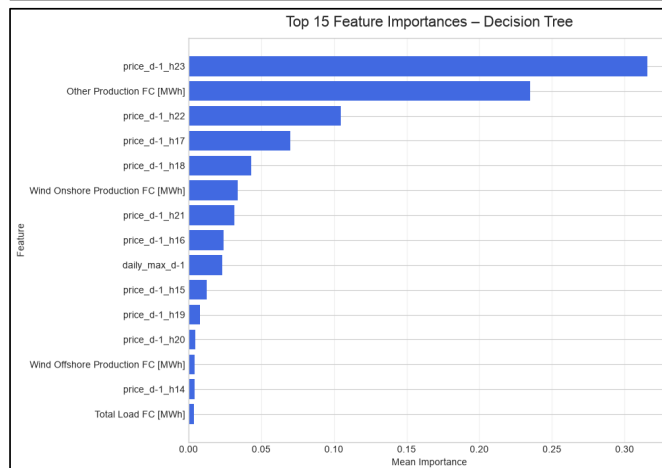
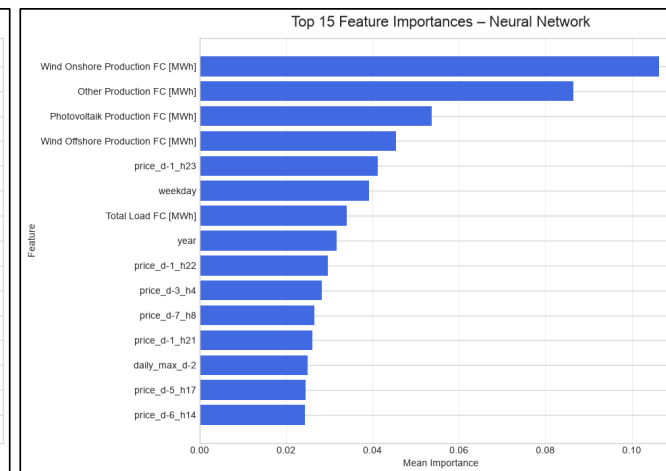
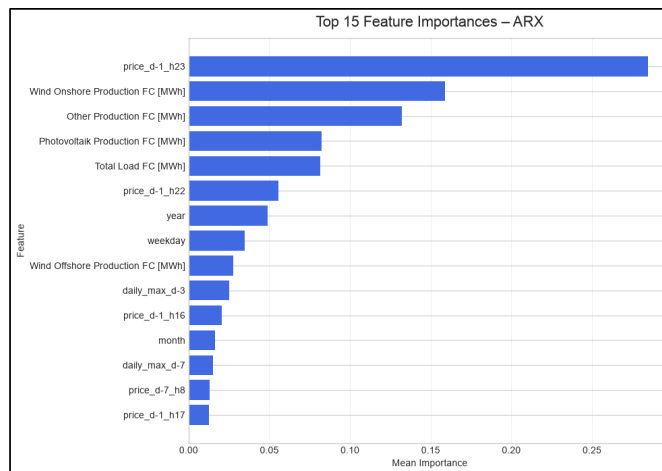


Back Up Graphics

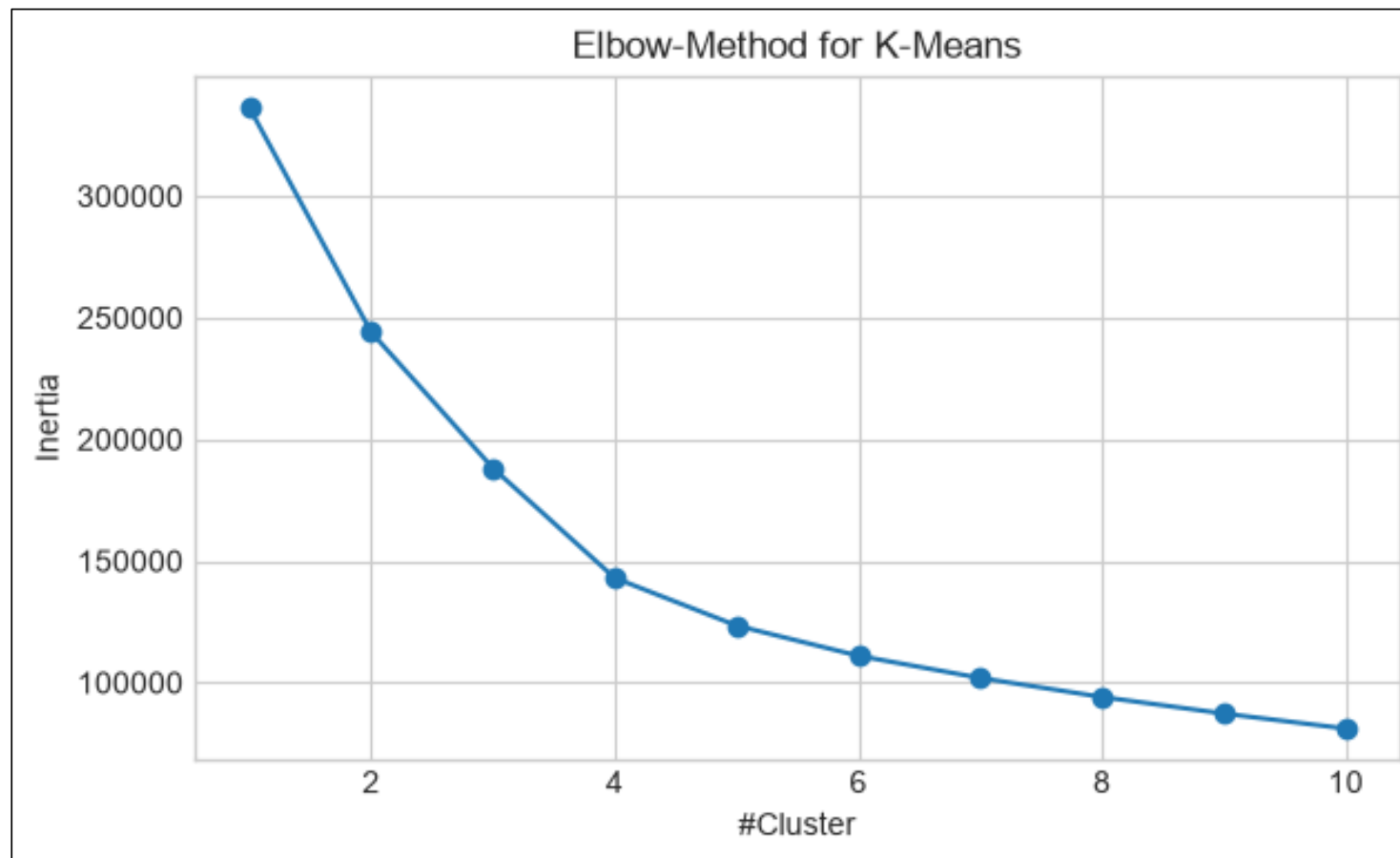
Actual vs. Predicted Day-Ahead Price per Model



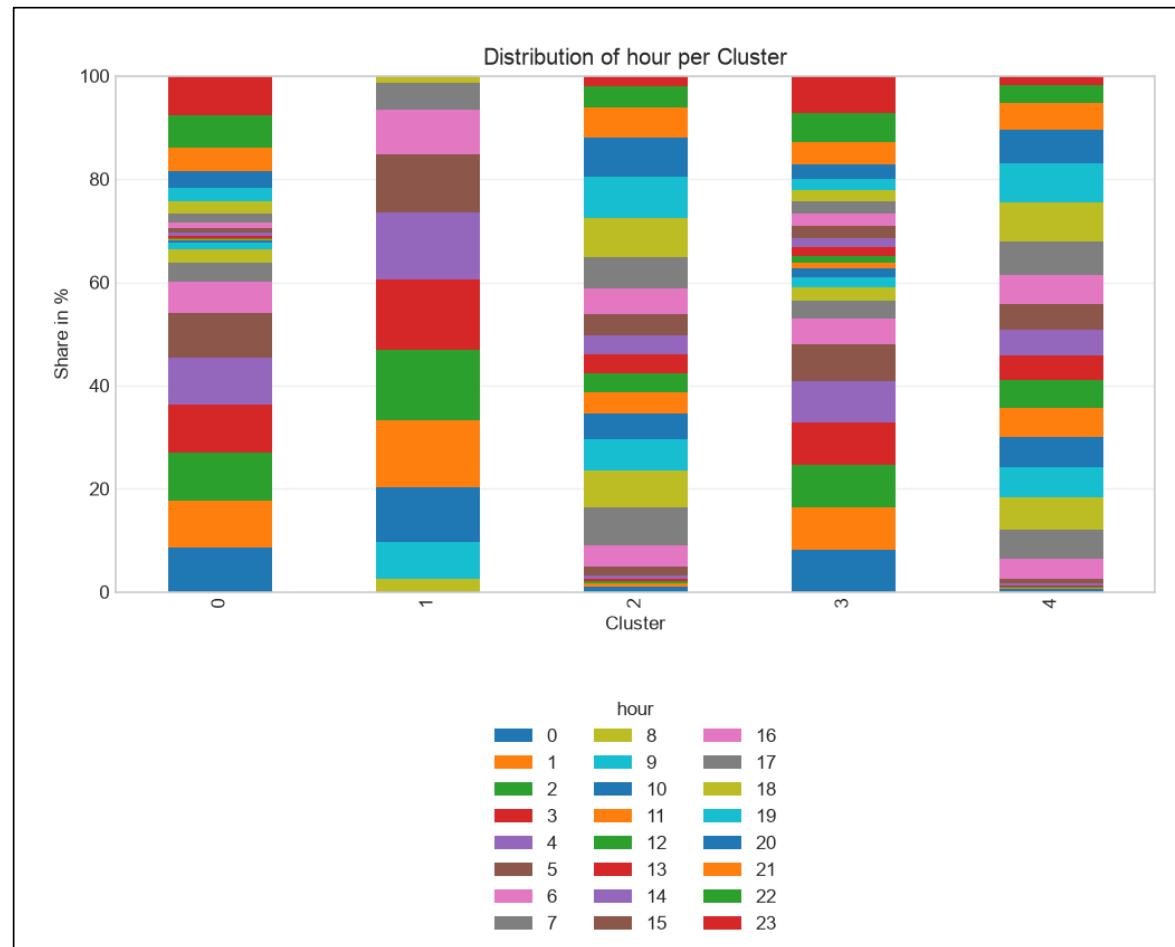
Top 15 feature importance per model



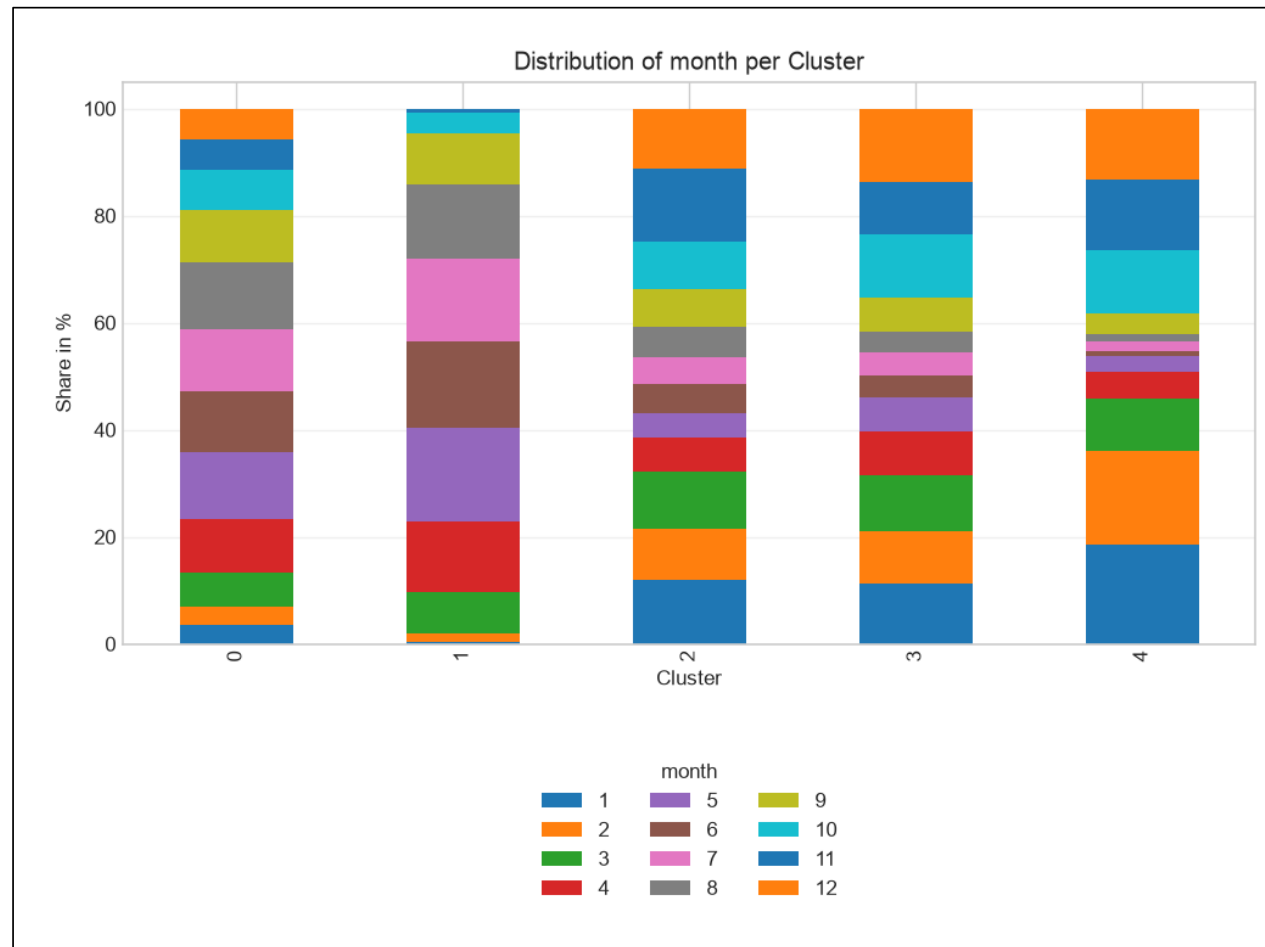
Elbow-Method for K-Means



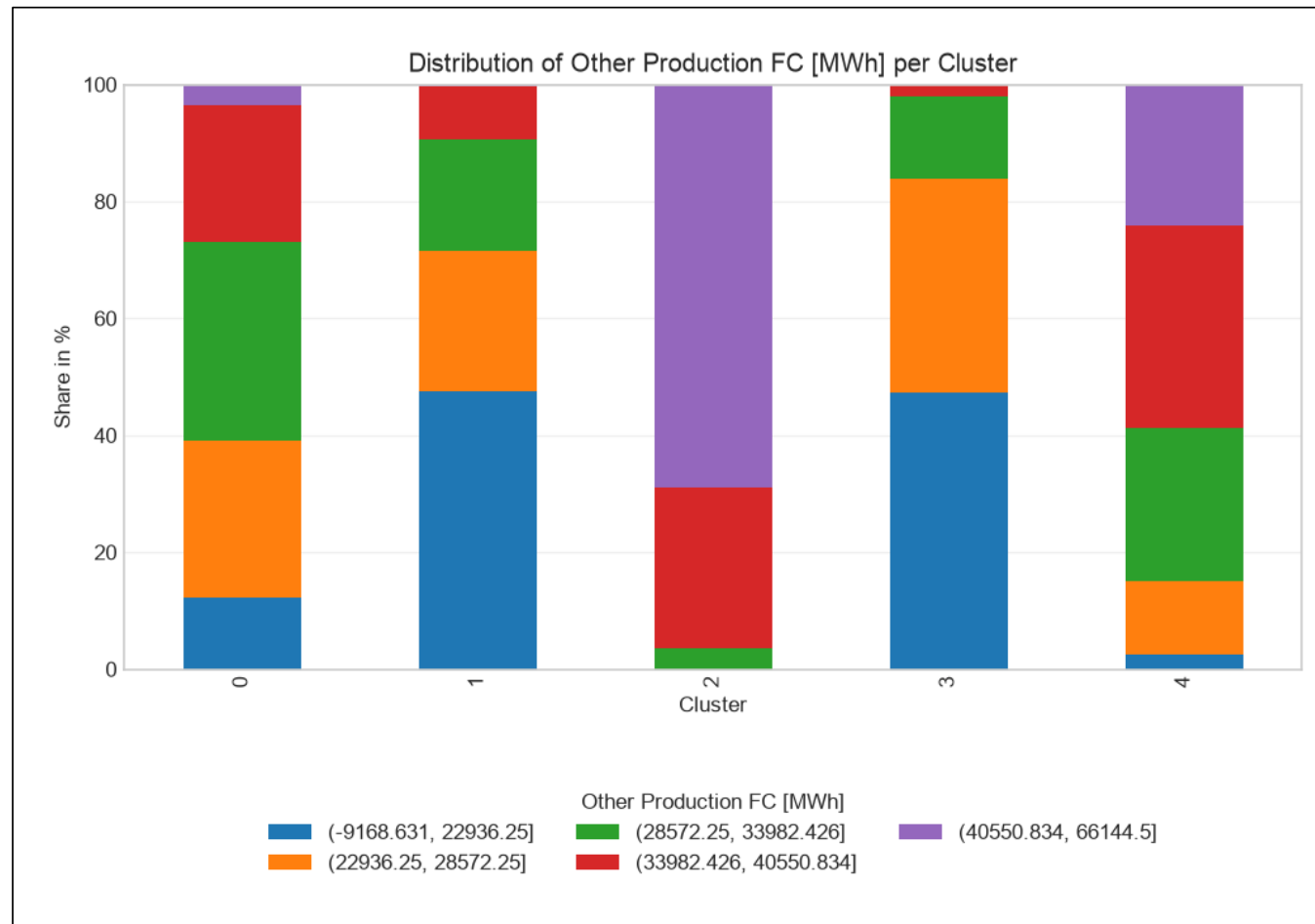
Distribution of hour per Cluster



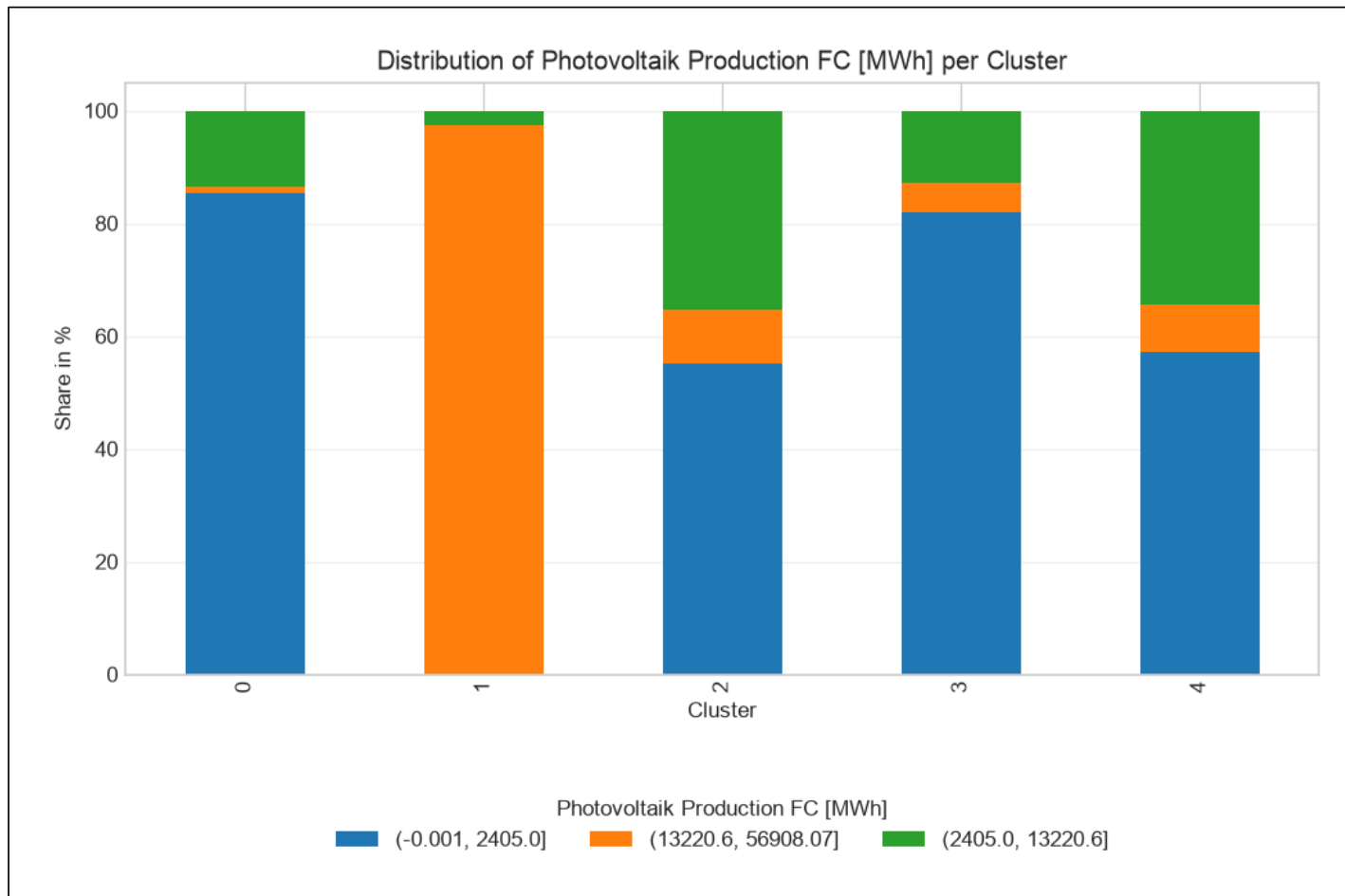
Distribution of month per Cluster



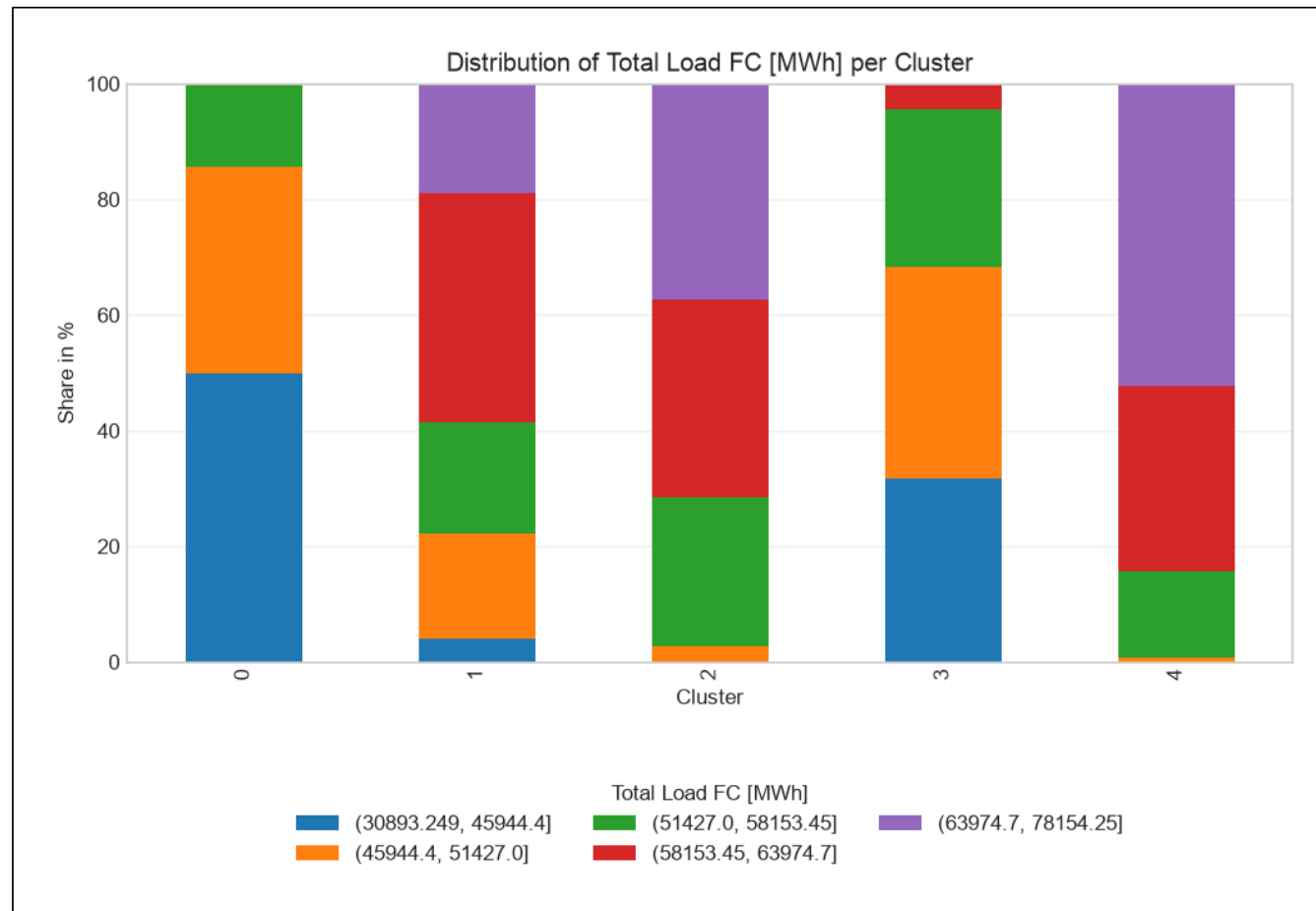
Distribution of Other Production FC per Cluster



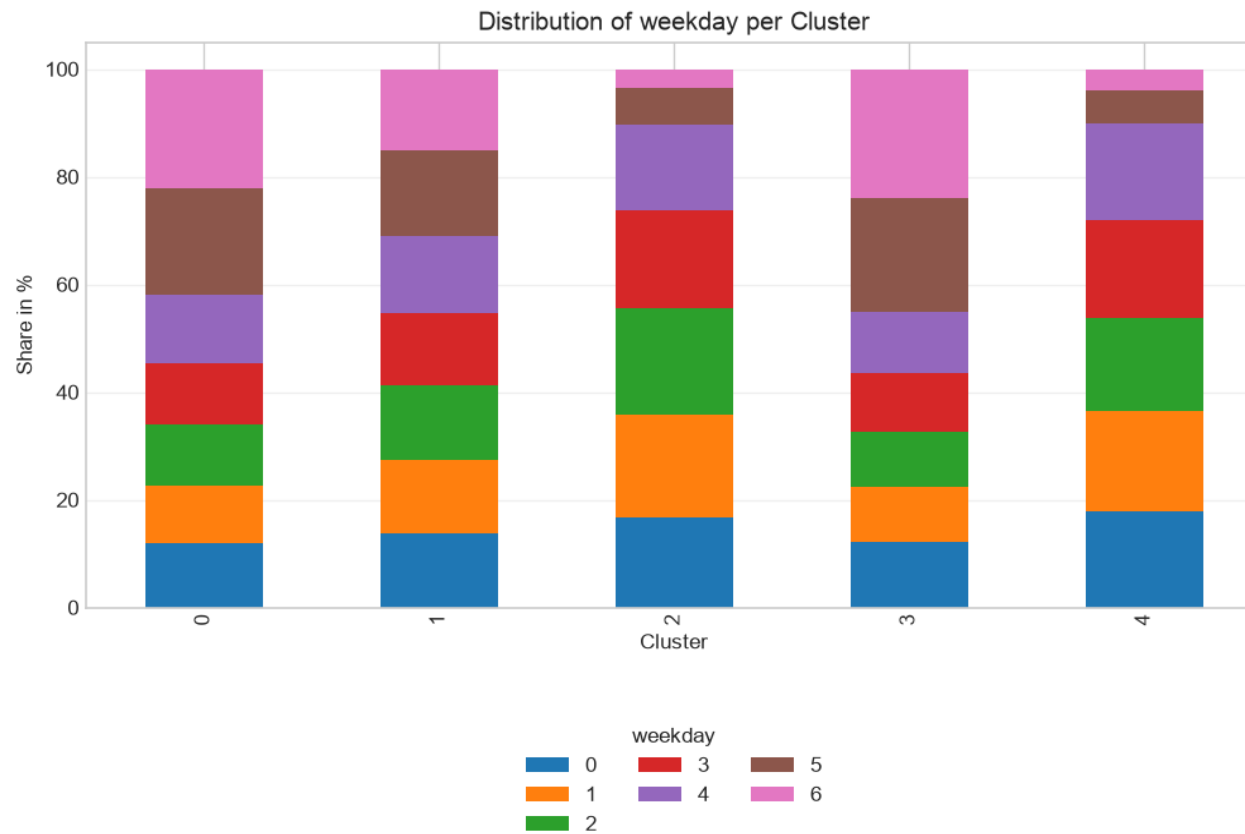
Distribution of Other Photovoltaik Production FC per Cluster



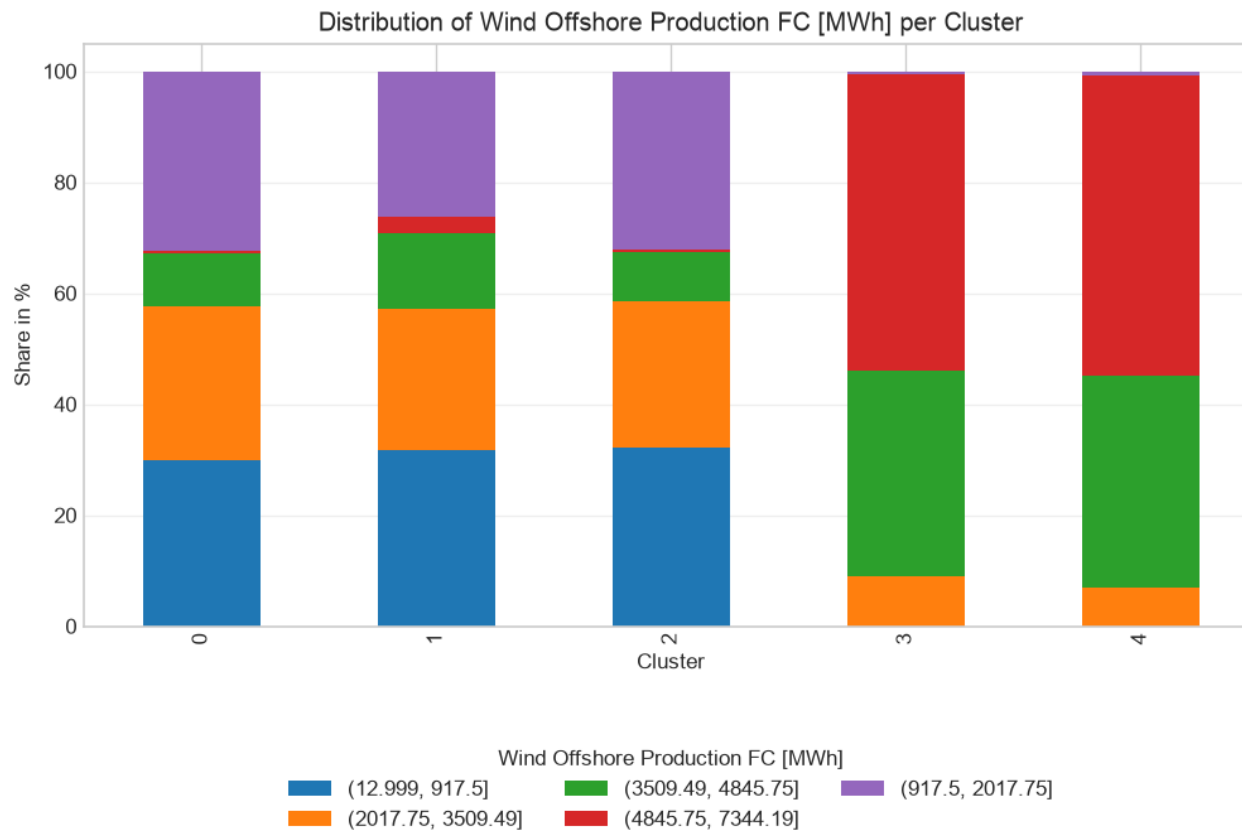
Distribution of Total Load FC per Cluster



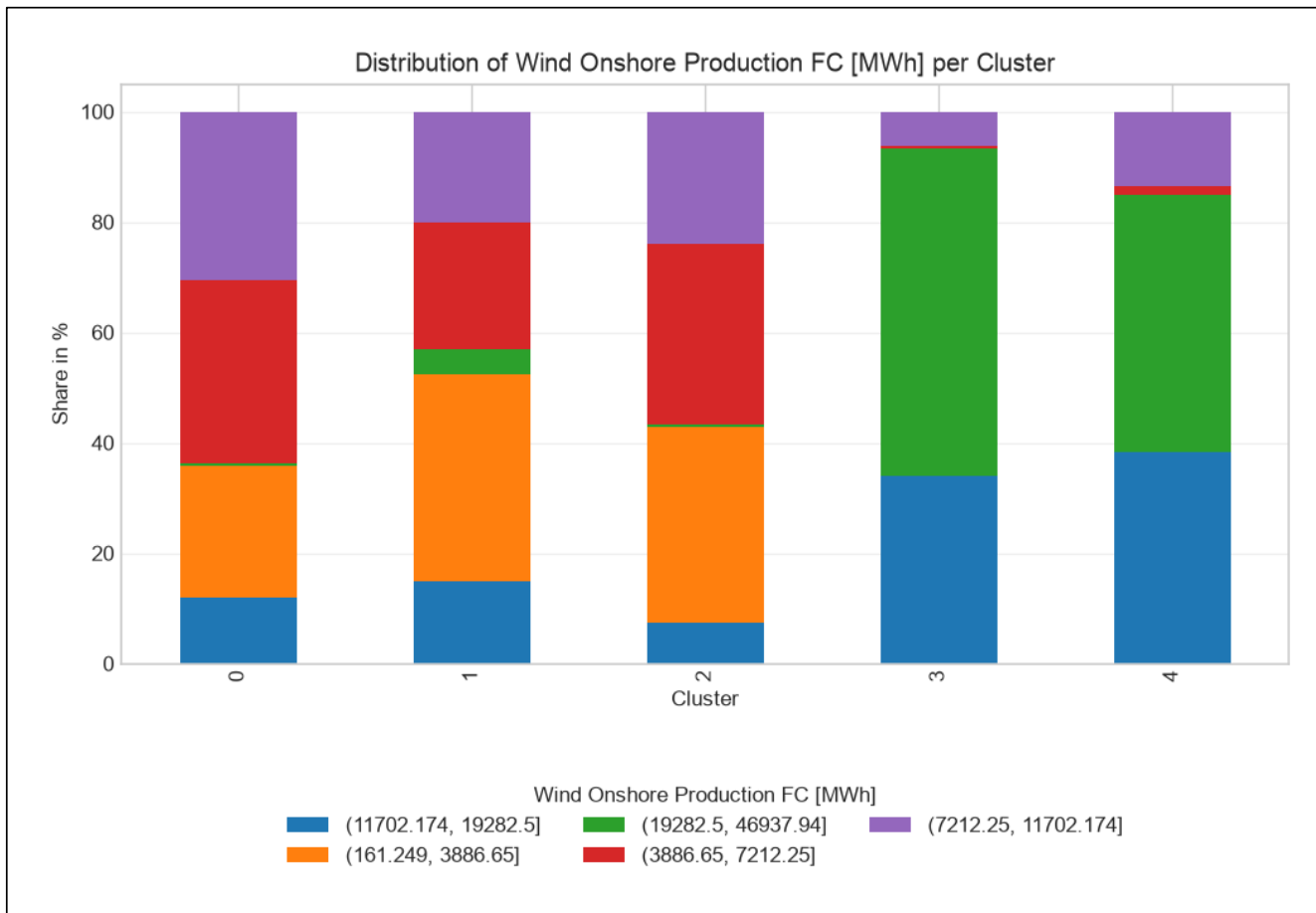
Distribution of weekday per Cluster



Distribution of Wind Offshore Production FC per Cluster



Distribution of Wind Onshore Production FC per Cluster



Distribution of Wind Onshore Production FC per Cluster

